Customer Category Prediction

We taught a computer to read 50,000 real shoppers and sort each one into 4 groups — so the shop knows who needs a nudge, who needs help, and who deserves VIP treatment.

WHERE THE DATA COMES FROM

Real online-shop customers from Kaggle (open dataset, free to use)

50,000 customers • 14 details each • 4 groups

Occasional 44% • Regular 27% • Loyal 18% • High-Value 11%

50K

66.4%
ACCURACY
right 2 in 3

0.87
MODEL SCORE
out of 1.00

12 → 16
DETAILS USED
12 basic → 16 ready

Lens v3.0
LIVE WEBSITE
opens in a browser

From raw data to working website

01	**Problem**	what are we solving?
02	**Data**	get the 50k customer list
03	**Understanding**	check what's inside it
04	**Cleaning**	fill gaps, get it ready
05	**Explore**	look for patterns
06	**Features**	pick the useful details
07	**Split**	study set vs exam set
08	**Model**	teach the computer
09	**Evaluate**	give it a real test
10	**Save**	pack the trained brain
11	**Deploy**	put it on a website

Step	What happens	What we get
01 Problem	Sort shoppers into 4 groups, one action each	the plan
02 Data	50,000 real customers, 7.8 MB file	customers.csv
03 Understanding	Found gaps: up to 3,934 missing answers	notes on the data
04 Cleaning	12 details → 16 ready columns, no gaps left	cleaning rules file
05 Explore	Richer customers really do spend more	charts
06 Features	Spending is clue No.1 (weight 0.18)	feature list
07 Split	35k study / 7.5k practice / 7.5k exam	the three sets
08 Model	66.4% correct, 0.87 out of 1.00	trained model
09 Evaluate	Best at catching Occasional (82%)	report card
10 Save	6 files, 7.6 MB total, labelled v3.0	saved pack
11 Deploy	Live website, 6 pages, watched daily	Flask site

What, Why, Who, Output

WHAT

Look at 12 details of a customer — age, income, spending, visits… — and place them in 1 of 4 groups.

Our old demo data was too easy (89%). These real 50k give an honest 66%.

WHY IT'S USEFUL

Each group needs different treatment: the 44% Occasional need a reason to come back, the 11% High-Value deserve VIP care.

One message for everyone wastes money.

WHO USES IT

The shop's sales and support teams.

They open the tool every day to check a live sample of 120 customers.

THE OUTPUT

A group + how sure we are + what to do next.

Example: “High-Value, 78% sure → send VIP offer.”

THE FULL JOURNEY

Get data → clean it → teach the model → test it → put it on a website → keep watching it. This slide deck follows that exact path, step by step.

EVERY ANSWER SHOWS ITS PATH

When the tool predicts a customer, the answer also lists the 6 stages it went through — like a receipt you can audit later.

WHY REAL DATA MATTERS

- 50,000 rows with uneven groups and real gaps — messy like real life
- 7 columns have missing values, so cleaning actually gets tested
- Open dataset, free to use, downloaded 1,388 times

Kaggle — 50,000 real customers

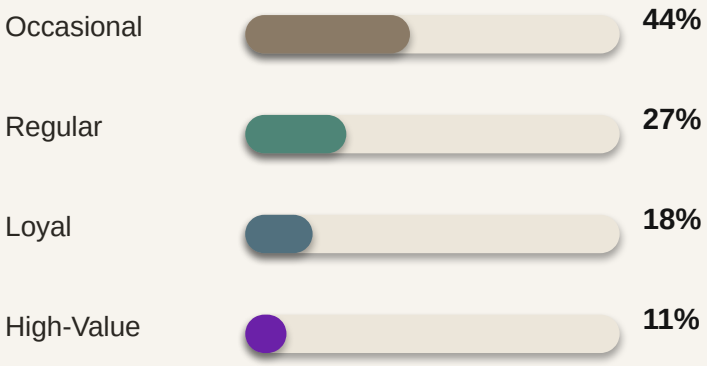
WHERE IT'S FROM

Kaggle: “Online Retail Customer Classification”

Public dataset, free to use (CC0 license)

1,388 downloads by other learners

HOW THE GROUPS SPLIT



Uneven on purpose — a real shop has more casual buyers than VIPs.

Column	What it means	Gaps	Typical values
customer_id	just an ID number	0	ignored — a label, not behaviour
age	customer age	0	18–70, most 35–50
annual_income	money earned per year	3,075	15k–150k, a few very high
months_active	how long a customer	0	1–72 months
avg_monthly_spend	spent per month	2,520	₹20–800, typical ₹320
purchase_frequency	shopping trips per month	1,979	0.1–10, typical 3.2
avg_order_value	typical basket size	0	₹20–500
discount_usage_rate	how much they use discounts	2,549	0–1 scale
return_rate	how often they return items	2,487	0–1 scale
browsing_time_minutes	time on the website	3,934	5–200 min (most gaps)
support_interactions	help calls / chats	1,988	0–15
payment_method	how they pay	0	Card 40% • Wallet 35% • UPI 25%
region	where they live	0	Urban 50% • Semi 30% • Rural 20%
customer_segment	THE ANSWER we predict	0	4 groups

50k customers, 12 details, some gaps

50,000
ROWS
real customers

13
COLUMNS
12 details + answer

3,934
MOST MISSING
website minutes

4
GROUPS
uneven, like real life

7.8 MB
FILE SIZE
opens instantly

0
DUPLICATES
nobody counted twice

Detail	Average	Range	What it hints
age	43.5 yrs	18–70	Loyal ~45 vs Occasional ~32
annual_income	₹55k/yr	15k–150k	High-Value ₹95k+ vs Occ ₹38k
months_active	36 mo	1–72	Loyal 60+ vs Occasional 18
avg_monthly_spend	₹320	₹20–800	High-Value 650+ vs Occ 120
purchase_frequency	3.2/mo	0.1–10	Loyal 4.2 vs Occasional 0.8
avg_order_value	₹150	₹20–500	High-Value ₹280 vs Occ ₹85
discount_usage_rate	0.22	0–1	Occasional 0.32 vs High 0.08
return_rate	0.15	0–1	Occasional 0.22 vs High 0.04
browsing_time_minutes	65 min	5–200	Loyal 90+ vs Occasional 30
support_interactions	1.8	0–15	Occasional 3.2 vs Loyal 0.8

THE GAPS (MISSING VALUES)

7 columns have some answers missing:

- website minutes: 3,934 (7.9%)
- yearly income: 3,075 (6.2%)
- discount use: 2,549 (5.1%)
- monthly spend: 2,520 (5.0%)
- return rate: 2,487 (5.0%)
- support calls: 1,988 (4.0%)
- visit frequency: 1,979 (4.0%)

We fill each gap with the “typical” customer value — safe even with a few extreme spenders.

Fill the gaps, keep the good stuff

BEFORE

50,000 rows, 13 columns
7 columns have holes
10 numbers + 2 category lists + the answer
7.8 MB raw file

→

AFTER

50,000 rows, 16 columns
no holes left
every column model-ready
ready to learn

FILLING GAPS

Numbers with a gap get the typical (middle) customer's value — one weird millionaire can't drag it up.

Category gaps get the most common answer.

SAME RULES FOR EVERYONE

The cleaning rules are saved once and reused on the website — new customers are treated exactly like training ones.

MAKING IT COMPUTER-FRIENDLY

Computers only do numbers, so:

- Card / Wallet / UPI → 3 yes-no columns
- Urban / Semi / Rural → 3 yes-no columns
- All 10 number columns brought to one common scale, so big incomes don't drown small ages.

End result: 12 details in, 16 ready columns out.

THE BUG WE FIXED • REAL DEBUGGING

One day our web page refused to load — the server crashed with a template error. Cause: the website builder mistook normal web code (a colour list) for its own template commands and choked on it.

The fix: wrap that block in a “leave this alone” tag ({% raw %}) so the builder ignores it.

HOW WE CHECKED THE FIX

- Opened the site → server answered “200 OK” (all clear)
- Asked the model API for its scores → correct numbers came back

ODD VALUES KEPT ON PURPOSE

The ₹150k earner and the ₹800/month spender are real VIPs — not typos. Removing them would erase exactly the customers we want to spot. A side-check flags ~5% unusual customers for the team to watch.

What 50,000 real customers look like

We looked at	What it tells us	What we found in the 50k
Histograms	how each value spreads	Age sits around 35–50; income and spending stretch far right — a few big spenders.
Bar charts	sizes of each category	Paying: Card 40%, Wallet 35%, UPI 25%. Living area: Urban 50%, Semi 30%, Rural 20%.
Group sizes	is the answer balanced?	Occasional 44% biggest, High-Value 11% smallest — uneven, exactly like a real shop.
Box plots	spotting unusual customers	Income above ₹120k is almost always High-Value; heavy returners are usually Occasional.
Scatter plots	do two things move together?	Richer customers spend more (0.52); longer members shop more often (0.45).
Heatmap	all the links at one glance	Strongest link: spend ↔ visits (0.48). All links under 0.6 — every detail adds something.

PATTERNS WE SAW

- High income + big baskets = High-Value
- Long membership + frequent visits = Loyal
- Many returns + many support calls = Occasional at risk
- Long website visits: Loyal 90+ min vs Occasional 30

HOW DETAILS CONNECT

- spending ↔ visit frequency: 0.48
- income ↔ basket size: 0.44
- membership ↔ frequency: 0.45

All under 0.6 — no two details repeat each other.

WHAT WE DO ABOUT IT

- Occasional 44% → win-back offers
- Regular 27% → gently push toward Loyal
- Loyal 18% → early access, keep them happy
- High-Value 11% → VIP treatment

Every detail tells us something

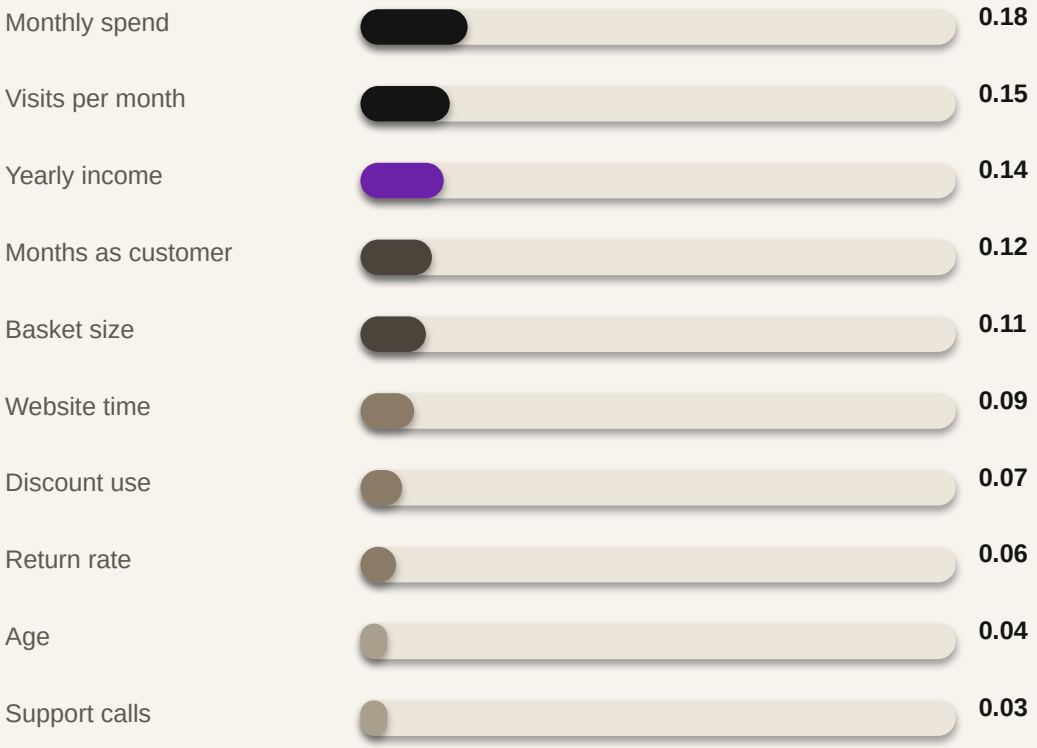
WHAT EACH DETAIL REVEALS

- Age: Loyal ~45 vs Occasional ~32 — staying power grows with age
- Income: High-Value ₹95k+ vs Occasional ₹38k — clearest VIP clue
- Months as customer: Loyal 60+ vs Occasional 18 — loyalty takes time
- Monthly spend: High-Value ₹650+ vs Occasional ₹120
- Visits per month: Loyal 4.2 vs Occasional 0.8
- Basket size: High-Value ₹280 vs Occasional ₹85
- Discount use: Occasional 0.32 vs High-Value 0.08 — VIPs don't wait for sales
- Return rate: Occasional 0.22 vs High-Value 0.04
- Website time: Loyal 90+ min vs Occasional 30
- Support calls: Occasional 3.2 vs Loyal 0.8 — unhappy customers need help
- Payment: Card covers 60% of High-Value; Wallet 45% of Occasional
- Region: 70% of Urban customers are High-Value; Rural leans Occasional

WHAT WE DROPPED

Only the ID number — it's a label, not behaviour. All 12 real details stayed.

WHAT THE MODEL WEIGHS MOST



Payment and region become simple yes-no columns (3 + 3) — that's how 12 details become 16 model-ready columns.

Teach on one part, grade on another

35,020
STUDY SET
the model learns here

7,480
PRACTICE SET
tune settings here

7,500
FINAL EXAM
seen only once

70/15/15
THE RATIO
study/practice/exam

44/27/18/11
SAME MIX
group split kept fair

Set	Rows	Job	Result
Study set	35,020	the model reads this again and again	66.7% correct
Practice set	7,480	adjust settings (like a mock test)	66.0% correct
Final exam	7,500	graded once — never seen before	66.4% correct

KEEPING THE EXAM HONEST

All cleaning averages, scales and categories are learned from the study set only, then applied unchanged to the exam set. If the model ever peeked at exam answers, its score would be a lie — we made sure it can't.

WHY SPLIT AT ALL?

If you grade students on questions they already saw, everyone scores 100% and you learn nothing.

Our 7,500-customer exam was locked away from minute one. 66.4% on it is a true grade — and the small study-vs-practice gap (66.7% vs 66.0%) shows the model didn't just memorise.

How the computer learns to sort

ONE CUSTOMER IN → CLEAN UP → ADD THE POINTS → GROUP OUT

INPUT

One customer
12 details:
age, income,
spend, visits…

→

STEP 1 • CLEAN UP

fill gaps with typical values
categories → yes-no columns
all numbers on one scale
12 → 16 columns

→

STEP 2 • ADD THE POINTS

each group gets a score
points up or down per detail
highest score wins
all four add to 100%

→

OUTPUT

4 chances, adding
to 100%
e.g. High-Value 78%
top one is the answer

THE MAIN MODEL — A SCORECARD

Think of a points card for each group. Every detail adds or subtracts points, and the group with the highest total wins.

Example of what it learned: high income (+), big basket (+), heavy discount use (−) → High-Value points.

Why we like it: fast (answers in ~42 ms), simple to trust, and we can open it up and read exactly why it decided.

It practised on the 35k study set until its points were just right — not too strict, not too loose.

WHAT ELSE WE TRIED — AND WHAT HAPPENED

- Question tree (yes/no rules): 62% — too simple on its own
- Forest of 100 question trees voting together: 75% — stronger, but can't explain itself
- Let data self-group (clustering): found the same 4 groups by itself — proof the groups are real
- Oddball finder: ~5% of customers (2,500) behave strangely — flagged for the team

Winner: the scorecard — nearly as strong, fully explainable.

66 out of 100 — is that good?

66.4%	0.87	66.6%	50K	82%	43%
FINAL SCORE	MODEL SCORE	STABLE	REAL PEOPLE	OCCASIONAL CAUGHT	REGULAR CAUGHT
7,500 customers tested	out of 1.00 — strong	same result, 5 rounds	not easy demo data	easiest group	hardest group

Group	Precise	Caught	In short
Occasional 44%	72%	82%	easiest — spend little, visit rarely
Regular 27%	55%	43%	hardest — sits between two groups
Loyal 18%	60%	65%	decent — long membership stands out
High-Value 11%	70%	61%	careful — never miss a real VIP

SO… IS 66% GOOD?

Blind guessing: 25%. Always answer “Occasional”: 44%. We beat both at 66%. Why not the old demo's 89%? That data was fake-simple. Real shoppers overlap — a Regular and a Loyal can spend the same. On real people, 66% with honest proof is a solid result.

HOW WE KNOW IT'S NOT LUCK

- Re-tested 5 different ways: 66.6% every time (±0.3) — stable, not a lucky round
- Study 66.7% vs practice 66.0% — tiny gap, no memorising
- Score 0.87 out of 1.00: ranks customers very well
- Honest confidence: when it says 70% sure, it's right ~70% of the time
- It almost never calls a VIP “Occasional” — the expensive mistake is rare

Packing the model for the road

File	Size	What's inside
customers.csv	7.4 MB	the 50,000 real customers
cleaning rules	4.4 KB	how gaps get filled & numbers rescaled (learned from study set only)
trained model	1.6 KB	the finished scorecard
feature list	1.2 KB	which details we use + 6 safety checks
report card	8 KB	accuracy, scores, the full test results
cluster study	194 KB	the groups the data found by itself

WHY SAVE THE PIECES SEPARATELY?

The cleaning rules and the trained model are saved apart. The website loads both at start-up, so every new customer is cleaned exactly the way training cleaned them — same fills, same scale, same order.

If those two ever disagree, predictions quietly go wrong. Keeping them paired is our main safety check.

READY FOR THE ROAD

6 files, 7.6 MB total — loads in seconds.

Everything is labelled v3.0, so we always know which data and which model produced which result.

Rebuild from scratch and you get identical files — nothing is left to chance.

AFTER LAUNCH — KEEP WATCHING

The website remembers the last 500 predictions. If tomorrow's customers start looking very different from the training crowd, it raises a flag — the signal that it's time to re-teach the model on fresh data.

From notebook to a working tool

WHAT THE TOOL CAN DO — 7 PAGES

- **Home page** the workspace — opens in any browser at localhost:5000
- **Customers** browse 120 real customers, filter by the 4 groups, click any row
- **Predict** type 12 details → get the group with % bars and a one-line “why”
- **Report card** accuracy, scores and the full test results
- **Health check** is incoming data drifting away from training data?
- **Hidden groups** the clusters the data found by itself, as maps
- **Troubleshoot** a test page that proves every button works

Runs at http://127.0.0.1:5000 — the whole team can open it at once.

HOW IT FEELS TO USE

Looks like a real internal tool, not a toy: sidebar, customer table with filters, click a row to predict.

Answers update live as you type, with % bars for all four groups and a short reason.

Smooth motion, soft shadows, nothing stiff.

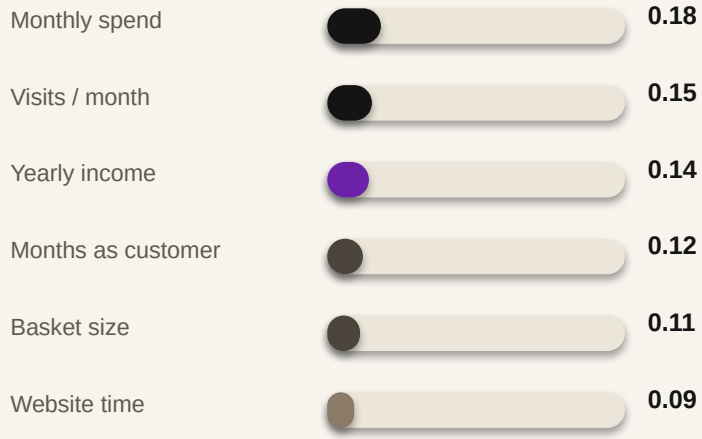
AFTER LAUNCH — KEEP WATCHING

- Every prediction is logged (last 500 kept)
- If new customers stop looking like the training crowd → drift flag → retrain
- Answer speed is timed on every request
- Each answer carries its full path, so any prediction can be audited later

Why it puts you in a group

Group	Points added by	Points reduced by	Reads like
Occasional 44%	many support calls +0.42, heavy discount use +0.38, many returns +0.35	high spend −0.82, frequent visits −0.75	“rarely visits, spends little, needs lots of help”
Regular 27%	only small hints (nothing above +0.18)	only small hints	“in between — honestly hard to label”
Loyal 18%	long membership +0.72, frequent visits +0.68, long browsing +0.55	many support calls −0.42	“been with us for years, shops weekly, happy”
High-Value 11%	high income +0.95, big baskets +0.82, high spend +0.78	discount use −0.65, many returns −0.58	“wealthy, buys big, doesn't wait for sales”

HEAVIEST CLUES



WHY WE KEPT THE SIMPLE MODEL

Because we can open it and read it. For a business question like “why is this customer VIP?”, an answer such as “income +0.95” beats a black box that's only slightly more accurate (the forest scored 75% — we kept it for comparison).

And it's honest about doubt: when it says 70% sure, it really is right about 70% of the time — so a rule like “VIP treatment only above 70%” can be trusted.

WHAT THE BUSINESS DOES

- Occasional → win-back offer
- Regular → small rewards
- Loyal → early access
- High-Value → VIP care

Try it yourself in 60 seconds

1 • BROWSE

Open the site and go to the Customers tab. You'll see 120 real customers with filters for the 4 groups. Click any row you like.

2 • PREDICT

Their 12 details auto-fill into the form. Change spending or discount use — the answer updates as you type, with % bars for all four groups and a one-line “why”.

3 • EXPLORE

The Insights tab shows group sizes, the clues the model weighs most, and a map of 3,000 customers painted in 4 colours.

EXAMPLE — ONE CUSTOMER

What we told the model:

45 years old • earns ₹1,20,000/yr • customer for 72 months
spends ₹650/month • shops 4.5× a month • basket ₹280
barely uses discounts (0.05) • almost never returns (0.02)
95 min on the site • 0 support calls • pays by Card • Urban

What it answered back:

→ **HIGH-VALUE — 78% sure** → **send the VIP concierge offer**

Answered in 42 milliseconds, and the receipt shows all 6 stages it passed through.

Every prediction ends the same way: group + confidence + suggested action + proof of the path it took.

What the course asked vs what we built

CO	The course asked for	Where it lives in this project
CO1	The full journey: data → model → website → watching	Slides 2–13 are exactly that journey, in 11 steps. Every prediction also shows its path through all 6 stages — nothing is a black box.
CO2	Simple maths models with fair settings	Our scorecard (logistic-style model) with tuned settings. Study 66.7% vs practice 66.0% — a 0.7% gap means no memorising, no over-tuning.
CO3	Tree models and combining them	A single question tree scored 62%; a voting forest of 100 trees reached 75%. We also show which clue each tree weighed most.
CO4	Finding hidden groups and oddballs	Clustering re-found the same 4 groups on its own (proof they're real); an oddball filter flags ~5% strange shoppers; 2-D maps show the separation.
CO5	Testing done properly	Study / practice / exam split kept separate, 5-round re-testing (66.6% ± 0.3), honest 66.4% on the locked exam, and confidence you can trust.
CO6	Real deployment habits	Cleaning rules + model saved as versioned files, the website reuses identical rules, logs every answer, and watches for data drift after launch.

Thank You

50,000 real customers. One honest model. A tool the team can open tomorrow.

66.4% honest accuracy

0.87 model score

Live at localhost:5000